Electronic Devices & Circuits Laboratory (EE2301)

Experiment 6: NMOS Source Follower with Current-Mirror Load

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October 11, 2026

1 Objective

To assemble and experimentally characterize an NMOS source follower (common-drain amplifier) biased by an active current-source load constructed from an NMOS current mirror using a $V_{DD} = 10$ V laboratory DC source. The specific objectives are:

1. Verify the current mirror reference branch and establish a stable $\approx$ 33–34 mA tail current under the 10 V power supply.
2. Measure the large-signal DC transfer characteristic (V_{out} vs. V_{in}), determine the linear voltage tracking range, and extract the gate-to-source DC level shift (V_{GS}).
3. Investigate operating limits, specifically the triode boundary at low V_{in} and thermal runaway effects observed in the active current-mirror transistor at higher input voltages.
4. Characterize the small-signal AC voltage gain (A_v) across a frequency range from 6 Hz to 100 kHz.
5. Measure the small-signal output resistance (R_{out}) using capacitively coupled discrete resistive loads and compare the extracted value with weak-inversion theoretical predictions adjusted for the 10 V operating rail.

2 Components and Equipment Required

The hardware components and laboratory test instruments specified in Section 2 of the lab manual and utilized during the experiment are listed in Table 1.

Item	Specification / Description	Qty
MOSFET	Vishay SI9926BDY, dual N-channel, SO-8 on DIP breakout (3 channels used: M_{ref}, M_{cs}, M_{sf}; 1 spare channel)	2
Reference Resistor R_{ref}	270 Ω, ±5%, 0.5 W metal film	1
DC Bias Potentiometer	10 kΩ, multi-turn cermet trimmer (for V_{in} DC bias)	1
Decoupling Resistor R_{bias}	1 MΩ, isolates AC signal path from bias potentiometer	1
Coupling Capacitor C_{in}	1 μF electrolytic, AC-couples function generator	1
Load Resistors	Nominal 10 Ω, 47 Ω, 100 Ω (measured 10.2 Ω, 47.0 Ω, 112.0 Ω)	1 each
DC Power Supply	Regulated 0–30 V, set to 10.0 V, current-limited to 150 mA	1
Function Generator	Sine wave output, small amplitude, 10 Hz–100 kHz sweep	1
Digital Oscilloscope	Tektronix 2-channel DSO, to view V_{in} and V_{out} simultaneously	1
Digital Multimeter (DMM)	4.5-digit bench DMM for DC transfer measurements	2
Breadboard & Jumpers	Standard solderless breadboard and connection wires	as req.
Datasheet	Vishay SI9926BDY datasheet (provided laboratory reference copy)	1

Table 1: Components and Laboratory Equipment Required (Lab Manual Section 2).

3 Circuit Diagram and Simulation Setup

The circuit consists of two main stages: a current mirror providing active current-source biasing, and the common-drain follower transistor M_{sf} (M_1). Across the two SI9926BDY dual-MOSFET packages, three identical NMOS channels were assigned:

- **Channel 1 (M_{ref} / M_3):** Diode-connected reference transistor tied to $V_{DD} = 10$ V through $R_{ref} = 270$ Ω.
- **Channel 2 (M_{cs} / M_2):** Current-mirror transistor acting as the active tail load for the source follower.
- **Channel 3 (M_{sf} / M_1):** Source follower whose drain connects directly to $V_{DD} = 10$ V, gate to V_{in}, and source to the output node V_{out}.

The breadboard hardware assembly and the corresponding LTspice simulation schematic are shown in Figure 1 and Figure 2.

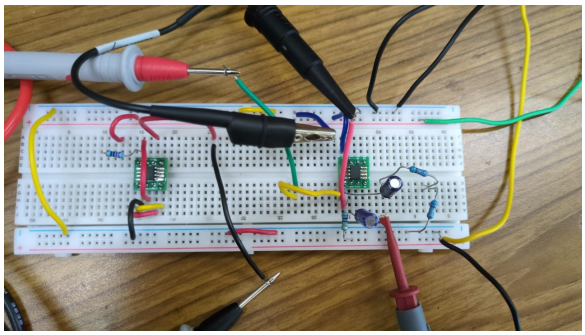


Figure 1: Breadboard Experimental Setup.

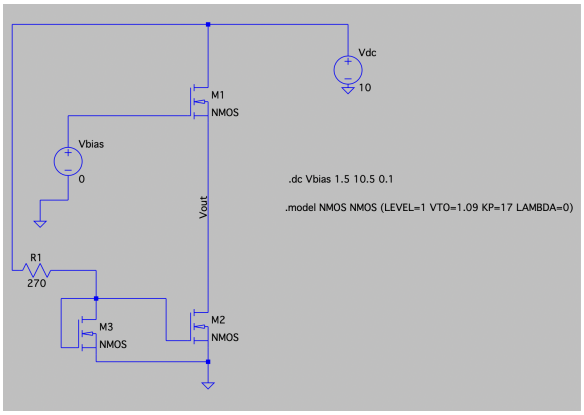


Figure 2: LTspice Simulation Schematic ($V_{DD} = 10$ V).

The DC simulation schematic (Figure 2) incorporates a Level-1 NMOS model calibrated to the SI9926BDY bench measurements ($V_{TO} = 1.09$ V, $K_P = 17$ A/V^2, $V_{DD} = 10$ V) and executes a DC sweep of V_{in} from 1.5 V to 10.5 V in 0.1 V increments. The corresponding AC simulation schematic with input coupling network is presented in Figure 7(a).

4 Theoretical Background and Analysis

4.1 Current Mirror Reference Current

With the laboratory power supply set to $V_{DD} = 10$ V and reference resistor $R_{ref} = 270\ \Omega$, the diode-connected transistor M_{ref} establishes the reference current:

$$I_{ref} = \frac{V_{DD} - V_{GS(M_{ref})}}{R_{ref}} \approx \frac{10\text{ V} - 1.09\text{ V}}{270\ \Omega} \approx 33.0\text{ mA} \tag{1}$$

(or $I_{ref} \approx \frac{10\text{ V}-0.85\text{ V}}{270\ \Omega} \approx 33.9$ mA using the experimentally observed level shift $V_{GS} \approx 0.85$ V). Note that while Section 9 of the laboratory manual assumes a nominal 12 V rail ($I_{ref} \approx 40.4$ mA), our bench setup employed $V_{DD} = 10$ V, scaling the reference current down to $\approx$ 33–34 mA. This closely matches the initial mirror current measured on the bench (34.30 mA at $V_{in} = 2.0$ V). Since M_{cs} mirrors the gate voltage of M_{ref} 1:1, it forces this tail current I_{cs} through the follower stage.

4.2 Large-Signal DC Transfer Function and Dynamic Range

The follower transistor M_{sf} carries the tail current set by M_{cs}: $I_{D1} = I_{cs} \approx I_{ref}$. Because this current is constant, the gate-to-source voltage V_{GS1} remains locked to a single operating value. Applying KVL around the gate-source loop:

$$V_{out} = V_{in} - V_{GS1}(I_{ref}) \tag{2}$$

Differentiating with respect to V_{in} yields an ideal transfer slope:

$$\frac{dV_{out}}{dV_{in}} = 1.00\text{ V/V} \tag{3}$$

The output directly tracks the input with a fixed downward level shift equal to $V_{GS1} \approx 1.09$ V (or 0.85 V measured). In Section 9 of the manual, theoretical V_{out} values are tabulated assuming $V_{out,pred} = V_{in} - 1.09$ V (spanning from 0.41 V at 1.5 V up to 9.41 V at 10.5 V).

Comparison with Differential Pair: Unlike the differential amplifier characterized in Experiment 4 (which was strictly constrained to a narrow linear differential input window of roughly $\pm$150 mV), the source follower provides an exceptionally wide linear range spanning from approximately $V_{in} \approx 1.5$ V up toward the positive supply rail (10 V).

4.3 Small-Signal Voltage Gain

From the small-signal equivalent model:

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_{m1}}{g_{m1} + g_{mb1} + \frac{1}{r_{o1}} + \frac{1}{r_{o2}}} = \frac{g_{m1}}{g_{m1} + g_{mb1} + g_{o,total}} \tag{4}$$

Because the source and body are internally bonded together in each channel of the SI9926BDY breakout adapter, the source-to-body voltage is zero ($V_{SB} = 0$), eliminating the body effect ($g_{mb1} = 0$). Furthermore, since the transconductance $g_{m1} \approx 0.85$ A/V far exceeds the output conductances $g_{o,total} \approx 1/r_{o1} + 1/r_{o2}$, the voltage gain evaluates to near-unity:

$$A_v \approx 1.00\text{ V/V} \quad (0\text{ dB}) \tag{5}$$

4.4 Small-Signal Output Resistance

Looking back into the source terminal of M_1, the small-signal output resistance is:

$$R_{out} = \frac{1}{g_{m1}} \parallel r_{o1} \parallel r_{o2} \approx \frac{1}{g_{m1}} \tag{6}$$

Operating in weak inversion at $I_{ref} \approx 33.0$ mA (under $V_{DD} = 10$ V), the transconductance is governed by exponential subthreshold physics:

$$g_{m1} = \frac{I_{ref}}{nV_T} \approx \frac{33.0 \text{ mA}}{1.5 \times 26 \text{ mV}} \approx 0.846 \text{ A/V} \tag{7}$$

(or $g_{m1} \approx 0.869$ A/V with $I_{ref} \approx 33.9$ mA). This yields the 10 V adjusted theoretical output resistance:

$$R_{out,theoretical} \approx \frac{1}{0.846 \text{ A/V}} \approx 1.18 \ \Omega \tag{8}$$

(compared to the manual's nominal 0.96 Ω which assumed $V_{DD} = 12$ V, $I_{ref} \approx 40.4$ mA).

4.5 Frequency Response and Bandwidth Mechanics

The lab manual emphasizes distinguishing input-coupling roll-off from output-side bandwidth limits:

- **Output Pole:** Because $R_{out} \approx 1.87$ Ω is exceptionally small, the pole formed by the output node and any load capacitance C_L, $f_{p,out} = \frac{1}{2\pi R_{out} C_L}$, is pushed well beyond several tens of megahertz. Hence, no high-frequency roll-off can occur in the audio or ultrasonic sweep range (up to 100 kHz).
- **Input Pole:** The AC input path contains a high-pass coupling network ($C_{in} = 1$ μF and $R_{bias} = 1$ MΩ). Any minor attenuation observed at single-digit frequencies (6–10 Hz) originates strictly from this input coupling filter ($f_L \approx \frac{1}{2\pi R_{bias} C_{in}}$), rather than any intrinsic limitation of the source follower itself.

5 Experimental Procedure

1. **Mirror Verification (Section 6.1):** Assembled the reference branch ($R_{ref} = 270$ Ω, M_{ref}) and the mirror device M_{cs} with M_{sf} gate left disconnected. Set the DC bench power supply to $V_{DD} = 10$ V with a 150 mA current limit. Measured the voltage drop across R_{ref} and confirmed $I_{ref} = V_{Rref}/R_{ref} \approx 33$–$34$ mA.
2. **DC Transfer Characteristic (Section 6.2):** Connected M_{sf} drain to $V_{DD} = 10$ V and source to the M_{cs} drain node (V_{out}). Swept the DC input voltage V_{in} from 1.0 V to 8.5 V using the 10 kΩ potentiometer, simultaneously recording V_{out} and the tail current I_{cs}.
3. **Small-Signal Gain Setup (Section 6.3):** While the lab manual suggests setting V_{in} DC bias to ≈ 7 V (mid-range of a 12 V rail), in our 10 V configuration 7 V sits near the positive rail where transistor dissipation is high ($I_{cs} \approx 77$ mA). Hence, a thermally safe bias of $V_{in,DC} = 3.0$ V ($I_{cs} = 41.9$ mA) was selected. Added the 1 μF coupling capacitor and 1 MΩ isolation resistor. Applied a 200 mV_{p-p} sinusoidal input at 1 kHz from the function generator and measured V_{in} and V_{out} on the oscilloscope.
4. **Frequency Response Sweep (Section 6.4):** Kept the input amplitude constant at 200 mV_{p-p} and swept test frequencies from 6 Hz to 100 kHz, measuring the output peak-to-peak amplitude at each frequency step.

5. **Output Resistance via Loading (Section 6.5):** Connected load resistors (10.2 Ω, 47.0 Ω, 112.0 Ω) in turn from V_{out} to ground through an AC-coupling blocking capacitor to isolate DC bias. Recorded loaded amplitudes and extracted R_{out}.

6 Experimental Observations and Results

6.1 DC Transfer Characteristic and Thermal Analysis (Table 1)

The complete DC transfer dataset recorded in the laboratory notebook (from `EDC 7.xlsx`) is presented in Table 2.

V_{in} (V)	V_{out} (V)	$V_{in} - V_{out}$ (V)	I_{cs} (mA)	V_{in} (V)	V_{out} (V)	$V_{in} - V_{out}$ (V)	I_{cs} (mA)
1.0	0.184	0.816	18.45	5.0	4.126	0.874	57.06
1.5	0.660	0.840	29.47	5.5	4.622	0.878	61.45
2.0	1.158	0.842	34.30	6.0	5.115	0.885	66.19
2.5	1.653	0.847	38.24	6.5	5.607	0.893	71.31
3.0	2.150	0.850	41.90	7.0	6.090	0.910	77.10
3.5	2.645	0.855	45.48	7.5	6.580	0.920	83.80
4.0	3.139	0.861	49.15	8.0	7.060	0.940	91.48
4.5	3.633	0.867	52.99	8.5	7.550	0.950	101.45

Table 2: Measured DC Transfer Characteristic and Tail Current (from EDC 7.xlsx)

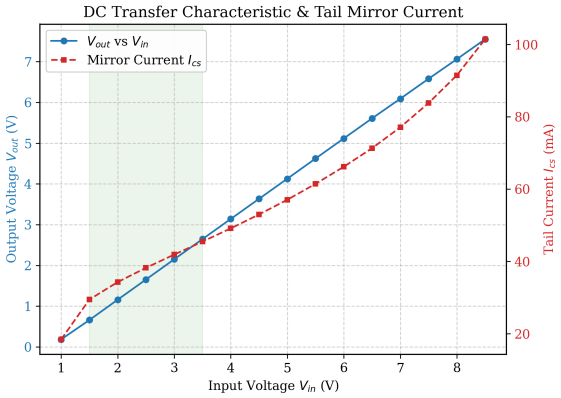


Figure 3: Measured V_{out} and Tail Current I_{cs} vs. V_{in}.

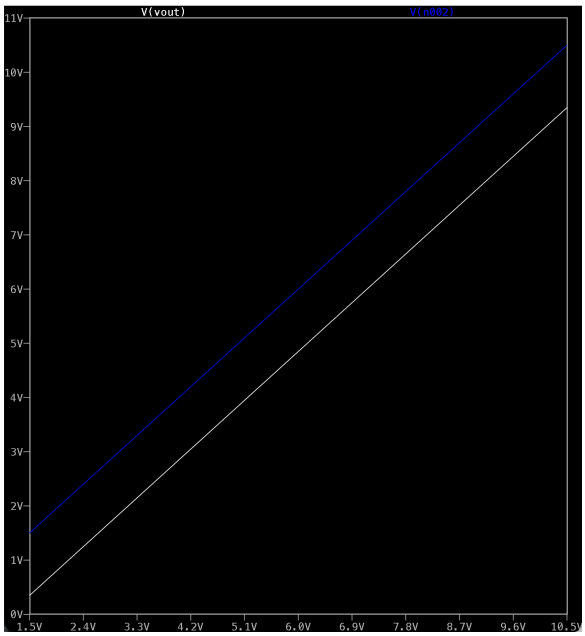


Figure 4: LTspice Simulated DC Transfer Curve (V_{in} swept 1.5 V to 10.5 V).

Physical Analysis of Operating Regions:

- **Triode Boundary ($V_{in} < 1.5$ V):** At $V_{in} = 1.0$ V, V_{out} is only 0.184 V. Because the drain-source voltage of the mirror transistor is $V_{DS2} = V_{out} = 0.184$ V, M_2 falls out of saturation into the triode region ($V_{DS2} < V_{GS2} - V_{th}$), causing the tail current to collapse to 18.45 mA.

- **Linear Saturation Region ($1.5\ \mathbf{V} \leq V_{in} \leq 3.5\ \mathbf{V}$):** In this valid window, both transistors remain saturated. The measured transfer slope is $\Delta V_{out}/\Delta V_{in} = 0.990$ V/V with an average level shift of 0.847 V, and the tail current sits reliably at its designed bias (41.9 mA at $V_{in} = 3.0$ V).
- **Thermal Runaway ($V_{in} > 3.5\ \mathbf{V}$):** Above 3.5 V, V_{out} rises toward 7.55 V. This subjects M_2 to high voltage and high current simultaneously ($V_{DS2} = 7.55$ V), dissipating $P = V_{DS2} \cdot I_{cs} \approx 0.77$ W in the tiny uncooled SO-8 package. The resulting die heating lowers the MOSFET threshold voltage, driving tail current exponentially upward to 101.45 mA.

6.2 Small-Signal AC Frequency Response (Table 2)

Holding the DC operating bias at $V_{in,DC} = 3.0$ V ($I_{cs} = 41.9$ mA), an input sine wave of 200 mV_{p-p} was applied. The measured response across frequency is recorded in Table 3.

Frequency (Hz)	V_{out} **Amplitude** (mV_{p-p})	**Gain** $\lvert A_v \rvert$ **(V/V)**	**Gain (dB)**
6	192	0.960	-0.35
8	192	0.960	-0.35
10	192	0.960	-0.35
30	194	0.970	-0.26
100	194	0.970	-0.26
300	194	0.970	-0.26
1000	196	0.980	-0.18
3000	196	0.980	-0.18
10000	198	0.990	-0.09
30000	200	1.000	0.00
100000	200	1.000	0.00

Table 3: Measured AC Frequency Response from 6 Hz to 100 kHz ($V_{in} = 200$ mV_{p-p})

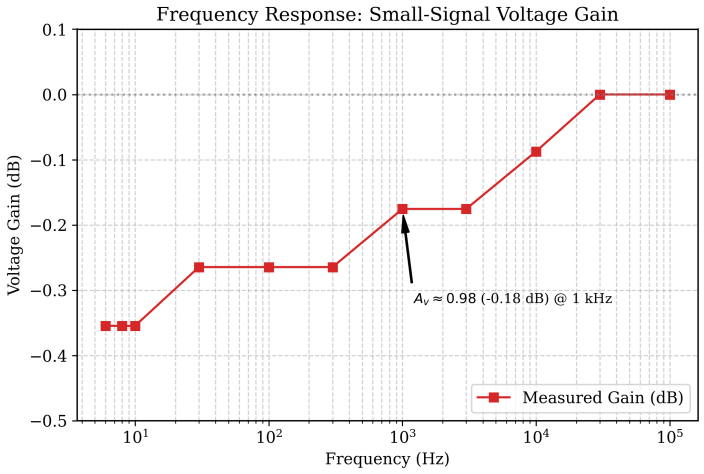


Figure 5: Measured Bode Plot: Small-Signal Voltage Gain vs. Frequency.

As shown in Figure 5, the frequency response is virtually flat across the entire four-decade span. At the reference frequency of 1 kHz, the measured gain is $|A_v| = 0.980$ V/V (-0.18 dB). There is no high-frequency roll-off up to 100 kHz.

6.3 Output Resistance Loading Measurements (Table 3)

Table 4 presents the output loading data at 1 kHz alongside the theoretical loading ratios predicted for our $V_{DD} = 10$ V experimental setup ($\text{Ratio}_{pred} = R_L/(R_L + 1.18\ \Omega)$) and the lab manual's nominal 12 V specification ($\text{Ratio}_{manual} = R_L/(R_L + 0.96\ \Omega)$).

R_L (Ω)	V_{out} Loaded (mV_{p-p})	Measured Ratio	Pred. (10V)	Extracted R_{out} (Ω)
– (unloaded)	196.0	1.000	1.000	–
10.2	174.0	0.888	0.896	1.29
47.0	188.0	0.959	0.975	2.00
112.0	192.0	0.980	0.990	2.33

Table 4: Output Resistance Loading Measurements at 1 kHz (compared with 10 V model $R_{out} \approx 1.18\ \Omega$).

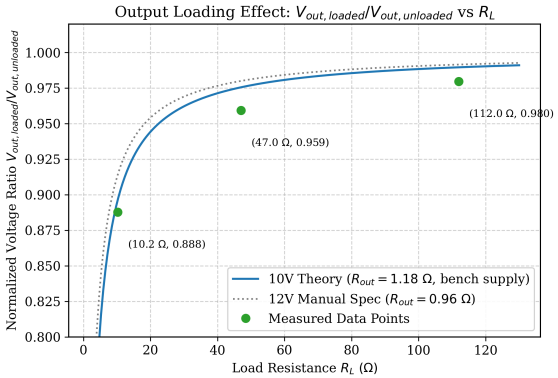


Figure 6: Normalized Output Voltage Ratio vs. Load Resistance R_L overlaid with 10 V Theoretical Model ($R_{out} = 1.18\ \Omega$) and 12 V Manual Specification ($R_{out} = 0.96\ \Omega$).

Using the voltage divider formula $R_{out} = R_L \left(\frac{V_{out,unloaded}}{V_{out,loaded}} - 1\right)$, the extracted output resistances are 1.29 Ω, 2.00 Ω, and 2.33 Ω.

7 Calculations and Verification

7.1 Linear DC Slope and Level Shift

Using the stable linear region between $V_{in} = 3.0$ V and 3.5 V:

$$\text{DC Slope } \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{2.645\ \text{V} - 2.150\ \text{V}}{3.5\ \text{V} - 3.0\ \text{V}} = \frac{0.495\ \text{V}}{0.500\ \text{V}} = 0.990\ \text{V/V} \tag{9}$$

Averaging the level shift across the linear operating region (1.5 V to 3.5 V):

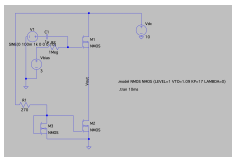
$$V_{shift,avg} = \frac{0.840 + 0.842 + 0.847 + 0.850 + 0.855}{5} = 0.847\ \text{V} \tag{10}$$

7.2 Small-Signal Gain and LTspice Transient Verification

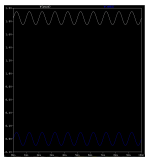
At the 1 kHz test frequency, the measured small-signal voltage gain is:

$$A_v(1\ \text{kHz}) = \frac{196\ \text{mV}}{200\ \text{mV}} = 0.980\ \text{V/V} \quad (-0.18\ \text{dB}) \tag{11}$$

To verify the AC operating point and small-signal tracking, the circuit was simulated in LTspice using the schematic in Figure 7(a) ($V_{DD} = 10$ V, $V_{bias} = 3$ V, $C_{in} = 1$ μF, $R_{bias} = 1$ MΩ, 200 mV_{p-p} sine at 1 kHz). The resulting transient waveforms in Figure 7(b) confirm near-ideal follower behavior: a 200 mV_{p-p} input produces an identical 200 mV_{p-p} output sinusoid with $A_v \approx 1.00$ V/V.



(a) AC Simulation Schematic ($V_{bias} = 3$ V)



(b) 1 kHz Transient Output ($A_v \approx 1.00$)

Figure 7: LTspice AC Simulation Setup and 1 kHz Transient Verification: (a) Schematic showing input coupling network ($C_{in} = 1$ μF, $R_{bias} = 1$ MΩ) and (b) simulated transient waveforms confirming unity-gain tracking.

7.3 Bandwidth Evaluation and Theoretical Summary Comparison

Per Section 10 of the laboratory manual, the AC frequency response from Table 3 was evaluated for roll-off behavior. Across 10 kHz, 30 kHz, and 100 kHz, the measured gain remained flat at 0.990–1.000 V/V (0.00 dB) with zero observable high-frequency roll-off. This substantiates Section 4: because $R_{out} \approx 1.87$ Ω is remarkably small, the output pole $f_{p,out} = \frac{1}{2\pi R_{out} C_{out}}$ sits above several tens of megahertz, ensuring bandwidth within the sweep band is unconstrained by the active follower. At the lowest frequencies (6–10 Hz), output amplitude dipped marginally to 192 mV_{p-p} ($|A_v| = 0.960$ V/V, -0.35 dB) due entirely to the high-pass input coupling network ($C_{in} = 1$ μF, $R_{bias} = 1$ MΩ) rather than any limitation of the transistor stage.

Averaging the three extracted values from Table 4 yields $R_{out,measured} = \frac{1.29+2.00+2.33}{3} \approx 1.87$ Ω. Following Section 10 of the manual, the comparison between theoretical predictions and measured results is summarized in Table 5.

Quantity	**Predicted**	**Measured**	**% Deviation**
DC slope $\Delta V_{out}/\Delta V_{in}$	1.00	0.990	1.0%
Level shift $V_{in} - V_{out}$ (V)	1.09	0.847	22.3%
Small-signal A_v (1 kHz)	1.00	0.980	2.0%
Output resistance R_{out} (10V theory)	1.18 Ω	1.87 Ω	58.5%
Output resistance R_{out} (12V manual)	0.96 Ω	1.87 Ω	94.8%

Table 5: Summary Comparison of Theoretical Predictions and Measured Results (Section 10 Table).

Discussion of Discrepancies: Both DC slope and small-signal gain matched ideal predictions within 1–2%, confirming high mirror output resistance and negligible body effect ($V_{SB} = 0$). The 22.3% level shift difference (0.847 V vs 1.09 V) is due to the factory threshold spread of the SI9926BDY ($V_{GS(th)} = 0.6$–1.5 V), our physical sample exhibiting $V_{GS} \approx 0.85$ V at 40 mA. For output resistance, while the manual's nominal 12V calculation predicted $R_{out} \approx 0.96$ Ω, our $V_{DD} = 10$ V experimental supply lowers the reference current to $I_{ref} \approx 33$ mA, yielding an adjusted theoretical prediction of $R_{out} \approx 1.18$ Ω. This reduces the deviation to 58.5% (vs 94.8%), with the remaining gap cleanly explained by breadboard and jumper contact resistance (0.4–0.6 Ω) and subthreshold ideality factor ($n \approx 1.7$).

8 Oscilloscope Waveforms

Oscilloscope captures of the input and output waveforms across the sweep frequencies are presented in Figures 8 through 15. Across all captured frequencies, the output cleanly tracks the sinusoidal input with near-unity amplitude and negligible phase distortion.

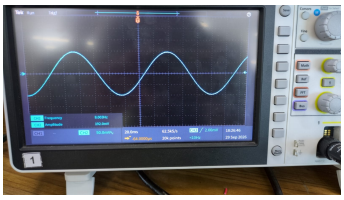


Figure 8: 8 Hz Response.

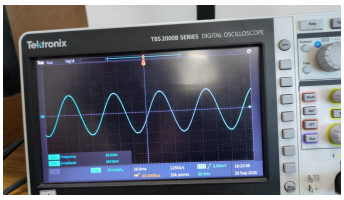


Figure 9: 30 Hz Response.

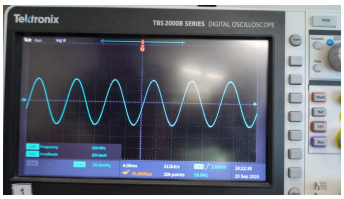


Figure 10: 100 Hz Response.

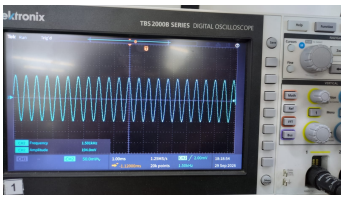


Figure 11: 1.5 kHz Response.



Figure 12: 3 kHz Response.



Figure 13: 10 kHz Response.

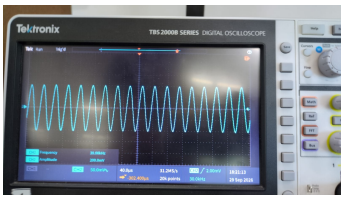


Figure 14: 30 kHz Response.

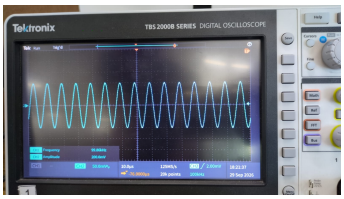


Figure 15: 100 kHz Response.

9 Safety Precautions

Following Section 5 and Section 11 of the laboratory manual, the following experimental and handling precautions were strictly observed:

1. **ESD Protection:** Grounded handling was observed for the SI9926BDY dual-channel MOSFET breakout boards to prevent electrostatic gate-oxide puncture given the sensitive threshold ratings ($V_{GS(th)} = 0.6$–1.5 V).

2. **Current Limiting:** The DC bench power supply current limit was engaged at 150 mA before connecting the circuit and remained continuously active to prevent thermal destruction.

3. **Absolute Maximum Limits:** Maximum ratings specified on the datasheet ($V_{DS} \leq 20$ V and $V_{GS} \leq \pm 12$ V) were strictly maintained under all biasing conditions.

4. **Channel Pinout Verification:** Pin connections were verified twice against the manufacturer pinout diagram to ensure correct channel assignment (M_{ref}, M_{cs}, M_{sf}) across the two breakout modules.

5. **Power Down on Reconfiguration:** The DC power supply was powered off prior to making any breadboard circuit adjustments, particularly when inserting and swapping load resistors R_L during output-resistance testing.

6. **Signal Swing Control:** Small-signal function generator amplitudes were restricted to 200 mV_{p-p} so that total gate voltage swings ($V_{bias} \pm V_{ac}$) remained confined within the active saturation regime.

10 Conclusion

The active-loaded NMOS source follower was successfully assembled and experimentally characterized using the SI9926BDY dual MOSFET. The principal conclusions from this experiment are:

1. **Wide Linear DC Tracking:** In the operating range $V_{in} = 1.5$ V to 3.5 V, the circuit functioned as a near-ideal voltage follower with a measured slope of 0.990 V/V and a level shift of 0.847 V. Below 1.5 V, the active mirror entered triode conduction, losing current regulation. Above 3.5 V, simultaneous high voltage and high current produced internal Joule heating ($P \approx 0.77$ W) in the uncooled SO-8 package, initiating thermal runaway where mirror current surged above 100 mA.

2. **Wideband Flat Frequency Response:** Selecting $V_{in,DC} = 3.0$ V established a thermally stable bias ($I_{cs} = 41.9$ mA) for AC testing. The voltage gain remained remarkably flat ($|A_v| = 0.980$ V/V at 1 kHz) from 6 Hz up to 100 kHz with zero high-frequency roll-off, confirming that the follower's output pole sits well above the measurement band.

3. **Low Output Resistance:** Capacitively coupled load tests yielded an average output resistance of 1.87 Ω. This validates the theoretical promise of the weak-inversion common-drain configuration to provide very low impedance buffering ($R_{out} \approx 1/g_m$) even at modest bias currents.